Assigment 4

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```
pharma<-read.csv("Pharmaceuticals.csv")
 row.names(pharma)<-pharma$Symbol
num<- pharma[, 3:11]
str(num)
```

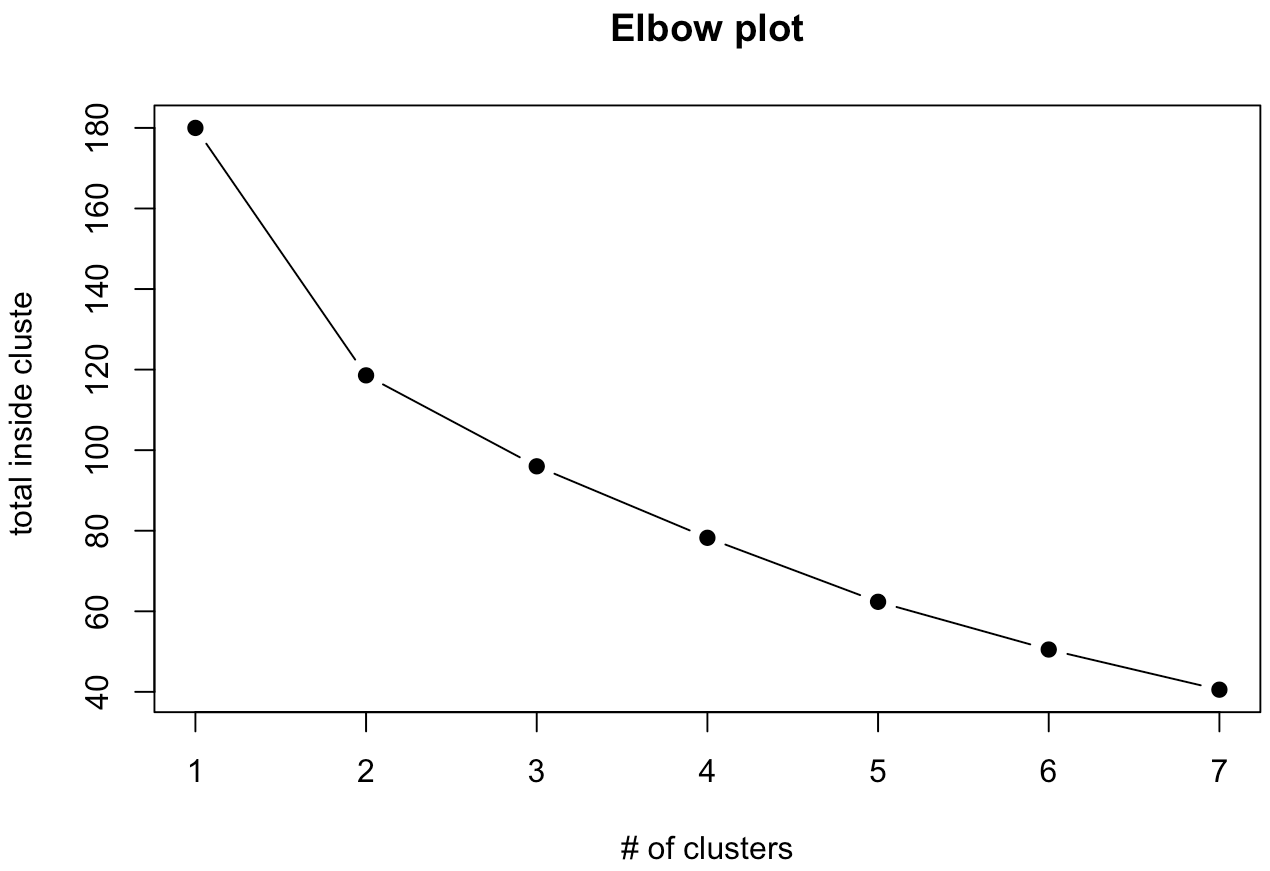
```
## 'data.frame':    21 obs. of  9 variables:
##  $ Market_Cap       : num  68.44 7.58 6.3 67.63 47.16 ...
##  $ Beta             : num  0.32 0.41 0.46 0.52 0.32 1.11 0.5 0.85 1.08 0.18 ...
##  $ PE_Ratio         : num  24.7 82.5 20.7 21.5 20.1 27.9 13.9 26 3.6 27.9 ...
##  $ ROE              : num  26.4 12.9 14.9 27.4 21.8 3.9 34.8 24.1 15.1 31 ...
##  $ ROA              : num  11.8 5.5 7.8 15.4 7.5 1.4 15.1 4.3 5.1 13.5 ...
##  $ Asset_Turnover   : num  0.7 0.9 0.9 0.9 0.6 0.6 0.9 0.6 0.3 0.6 ...
##  $ Leverage         : num  0.42 0.6 0.27 0 0.34 0 0.57 3.51 1.07 0.53 ...
##  $ Rev_Growth       : num  7.54 9.16 7.05 15 26.81 ...
##  $ Net_Profit_Margin: num  16.1 5.5 11.2 18 12.9 2.6 20.6 7.5 13.3 23.4 ...
```

```
library(cluster)
z <- scale(num)
round(apply(z, 2, sd), 2)
```

```
##        Market_Cap              Beta          PE_Ratio               ROE
##                 1                 1                 1                 1
##               ROA    Asset_Turnover          Leverage        Rev_Growth
##                 1                 1                 1                 1
## Net_Profit_Margin
##                 1
```

#a. k means gets its data by distance so any large numbers would causle an outlier. I made everything have an equal variable whieght so that nothing cause z scores or k- means to cause a skew in the data.

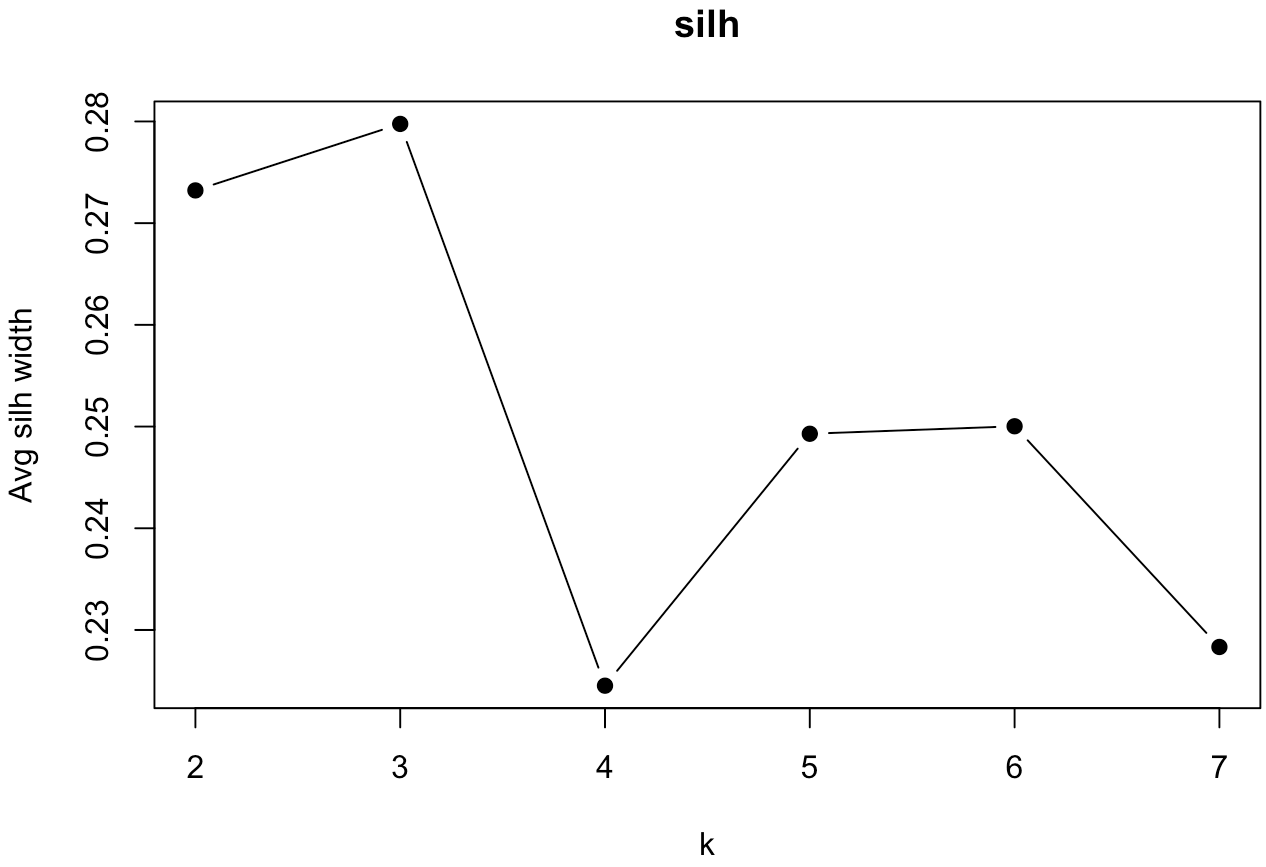
```
set.seed(1)
clustersum <-sapply(1:7, function(k) kmeans(z, centers = k, nstart=50)$tot.withinss)
plot(1:7, clustersum, type ="b", pch = 19,
     xlab = "# of clusters", ylab ="total inside cluste ",
     main= "Elbow plot")
```




```
d <- dist(z)
set.seed(1)
avg<- sapply(2:7, function(k){
  Kmen<- kmeans(z, centers = k,nstart =51)
  mean(silhouette(Kmen$cluster, d)[, "sil_width"])
})
```

THe plot shows a downward slop that is conststent across. it shows a decrease as you increase clustes.

```
plot(2:7, avg, type ="b", pch = 19,
xlab="k", ylab ="Avg silh width", main ="silh")
```




```
round(avg,3)
```

```
## [1] 0.273 0.280 0.225 0.249 0.250 0.228
```

for k I choose 3 because it had the highest average shiloute score

```
set.seed(1)
kMS<- kmeans(z, centers = 3, nstart =51)
kMS$size
```

```
## [1]  4 11  6
```

```
pharma$Cluster<- kMS$cluster
split(rownames(pharma),pharma$Cluster)
```

```
## $`1`
## [1] "AGN" "AHM" "BAY" "PHA"
##
## $`2`
##  [1] "ABT" "AZN" "BMY" "LLY" "GSK" "JNJ" "MRK" "NVS" "PFE" "SGP" "WYE"
##
## $`3`
## [1] "AVE"  "CHTT" "ELN"  "IVX"  "MRX"  "WPI"
```

```
profile <-aggregate(num, by = list(Cluster = kMS$cluster), FUN = mean)
round(profile, 2)
```

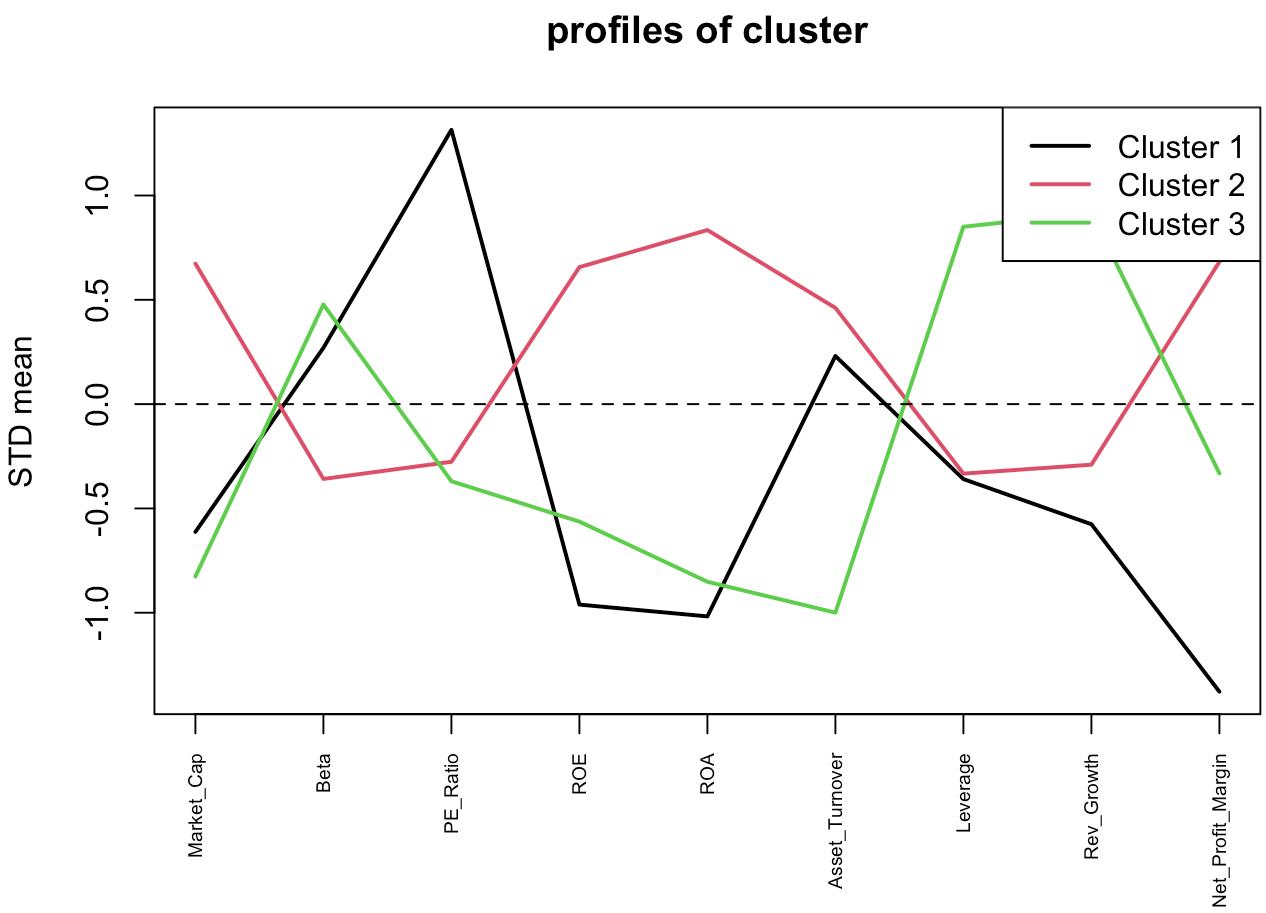
```
##   Cluster Market_Cap Beta PE_Ratio  ROE   ROA Asset_Turnover Leverage
## 1       1      21.75 0.60    46.90 11.3  5.10           0.75     0.30
## 2       2      97.11 0.43    20.95 35.7 14.95           0.80     0.33
## 3       3       9.23 0.65    19.43 17.3  5.98           0.48     1.25
##   Rev_Growth Net_Profit_Margin
## 1       7.01              6.65
## 2      10.16             20.17
## 3      23.49             13.52
```

```
round(kMS$centers, 2)
```

```
##   Market_Cap  Beta PE_Ratio   ROE   ROA Asset_Turnover Leverage Rev_Growth
## 1      -0.61  0.27     1.31 -0.96 -1.02           0.23    -0.36      -0.58
## 2       0.67 -0.36    -0.28  0.66  0.83           0.46    -0.33      -0.29
## 3      -0.83  0.48    -0.37 -0.56 -0.85          -1.00     0.85       0.92
##   Net_Profit_Margin
## 1             -1.38
## 2              0.68
## 3             -0.33
```

#bCLuster 1 market cap in the middle at 21.75 pe is the largest at 46.9. roe and roa are low. CLuster 2 has the larget cap at 97.1 The higest roe and roa and net profit of 20. THis company seems to be the most profitable # cluster 3 is the middle it has the smallest market cap and highest beta scor this apperas to be a small biz.

```
matplot(t(kMS$centers), type = "l", lty= 1, lwd =2, col = 1:3, xaxt ="n",
        ylab = "STD mean", main = "profiles of cluster")
axis(1, at = 1:9, labels = colnames(num), las =2, cex.axis =.6)
legend("topright", legend = paste("Cluster", 1:3), col =1:4, lty =1, lwd =2)
       abline(h=0, lty = 2)
```




```
table(pharma$Cluster, pharma$Median_Recommendation)
```

```
##
##     Hold Moderate Buy Moderate Sell Strong Buy
##   1    2            1             0          1
##   2    6            3             2          0
##   3    1            3             2          0
```

```
table(pharma$Cluster , pharma$Location)
```

```
##
##     CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
##   1      1      0       1       0           0  1  1
##   2      0      0       0       0           1  2  8
##   3      0      1       0       1           0  0  4
```

```
table(pharma$Cluster, pharma$Exchange)
```

```
##
##      AMEX NASDAQ NYSE
##   1    0      0    4
##   2    0      0   11
##   3    1      1    4
```

#c cluster 1 in cluse canda,germant uk and us. Cluster 2 had the most state side at 8, and Switz and Uk. Cluster 3 had france, ireland and us. Cluster 1 and 2 had all NYSE stocks. CLsuter 3 was more ddiverse at amex and nasdaq and NYSE.This shows that accross the board things can change depending on comapnies outlooks

D cluster names cluster one is companies that cost alot- the withb. highest p/e score

#2 high profitable companies big companies that make a lopt of money #3 Small companies with potetial for quick growth across the board due to high grow rev score. # to conclude we put 21 pharms into 3 clusters based on differnt sizes profits and growth margins and more,